MONITORING K8S: FOLLOW THE DATA

DANIEL MAHER, DATADOG

@phrawzty

1. TRENDS IN CONTAINERISATION

2. THE QUALITIES OF GOOD METRICS

3. SCALING KUBERNETES (W/ METRICS)

DANIEL MAHER

DOCS & TALKS
DEVOPSDAYS
GREAT OUTDOORS
GOURMAND
SUITS FOR NO REASON

@phrawzty

DATADOG

SAAS-BASED MONITORING

TRILLIONS OF POINTS/DAY

WE'RE HIRING:
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DATADOG

Docker support #844

Merged **remh** merged 14 commits into `DataDog:master` from `steeve:master` on Mar 8, 2014

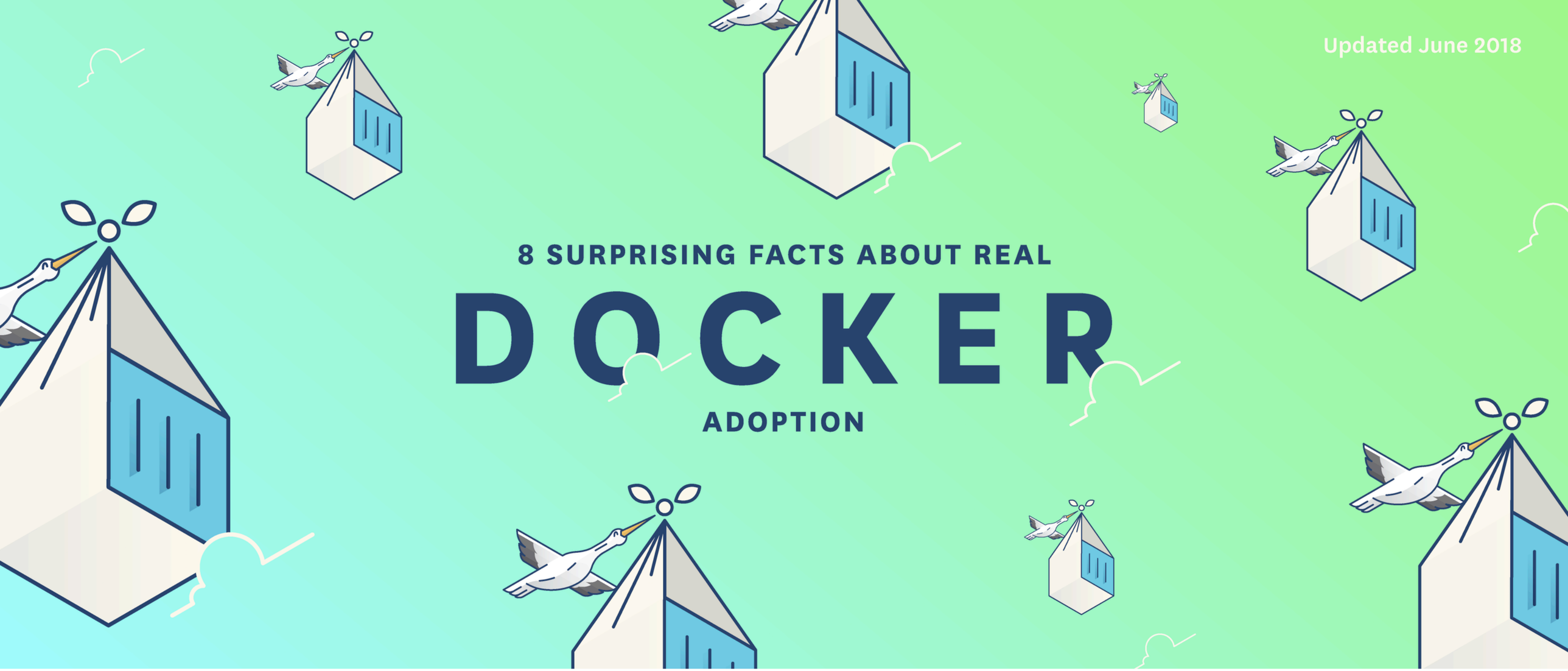
Conversation 23 | Commits 14 | Files changed 2

steeve commented on Feb 25, 2014

This PR enables instrumentation on Docker (LXC/cgroups) containers for:

- Memory
- CPU
- Disk

It uses the native Unix socket to communicate with `dockerd`, and thus doesn't need any



https://www.datadoghq.com/docker-adoption/

https://www.datadoghq.com/container-orchestration/

Docker Adoption

Dabbler: used Docker during a given month, but hadn’t reached significant use as defined by Adopter.

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Adopter: the average number of containers running during the month was at least 50% the number of distinct hosts run, or there were at least as many distinct containers as distinct hosts run during the month.

Docker Adoption

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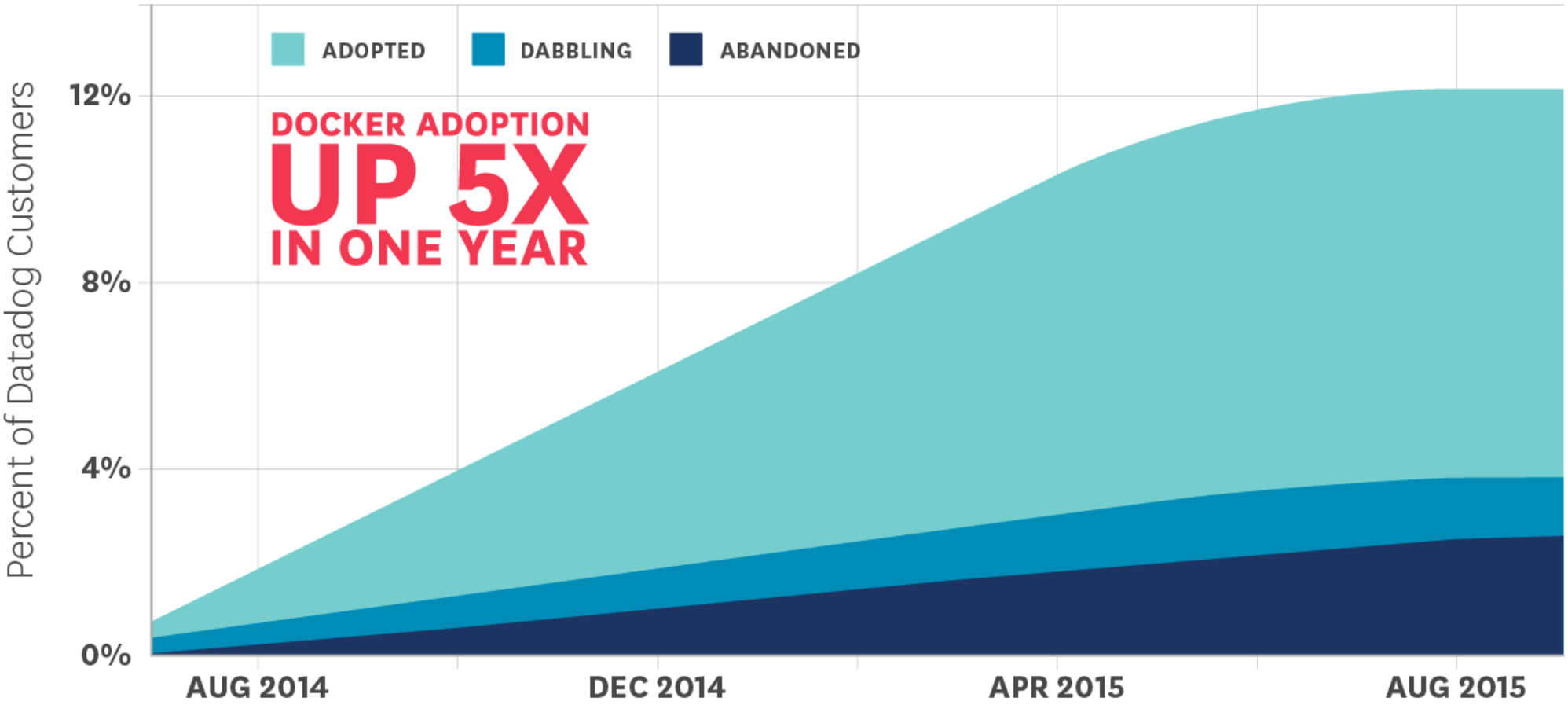
Adopter: the average number of containers running during the month was at least 50% the number of distinct hosts run, or there were at least as many distinct containers as distinct hosts run during the month.

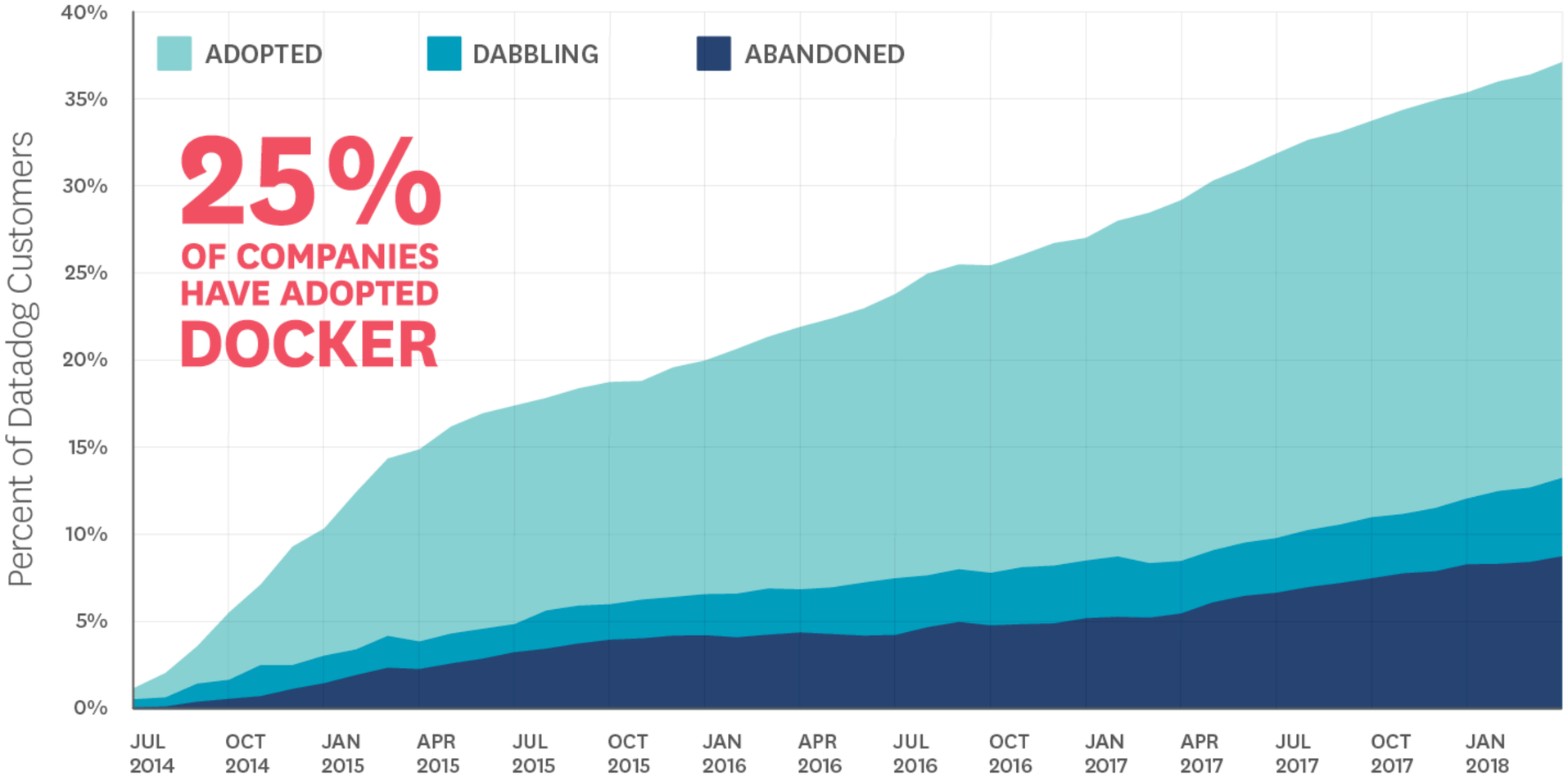
Abandoner: a currently active company that used Docker in the past, but hasn't used it at all in the last month.

Docker Adoption Behavior



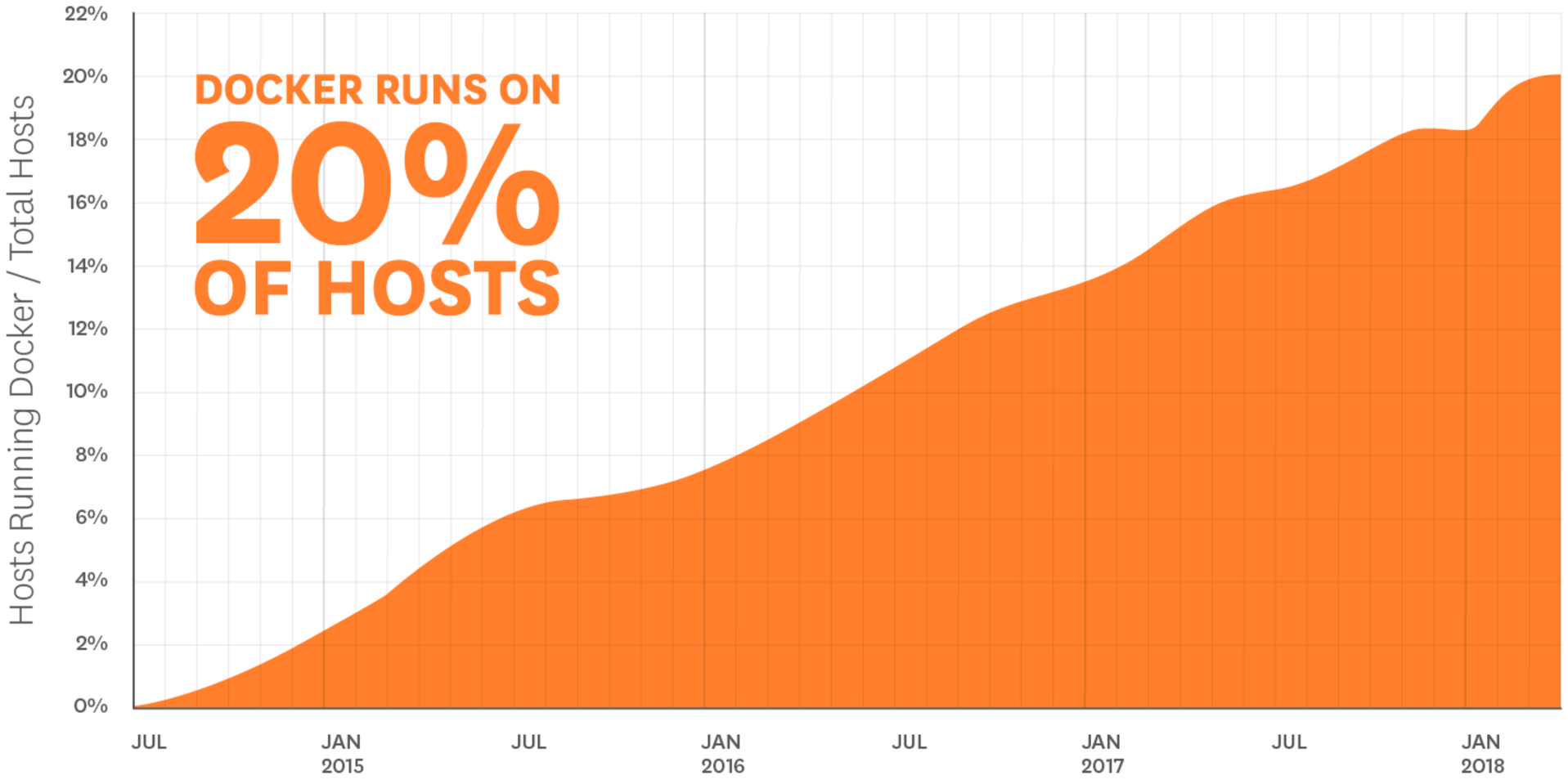
Source: Datadog

Docker Adoption Behavior



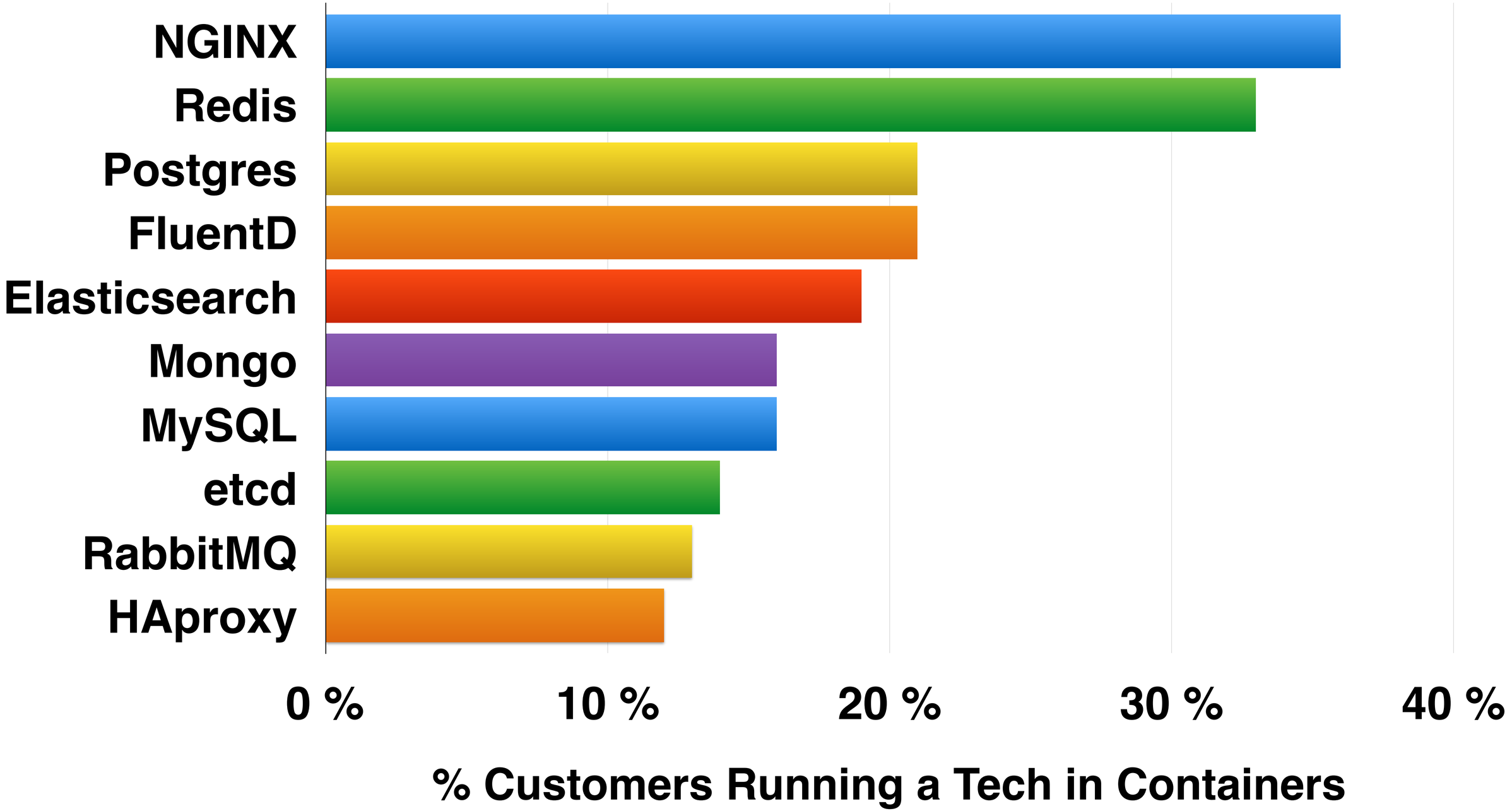
Source: Datadog

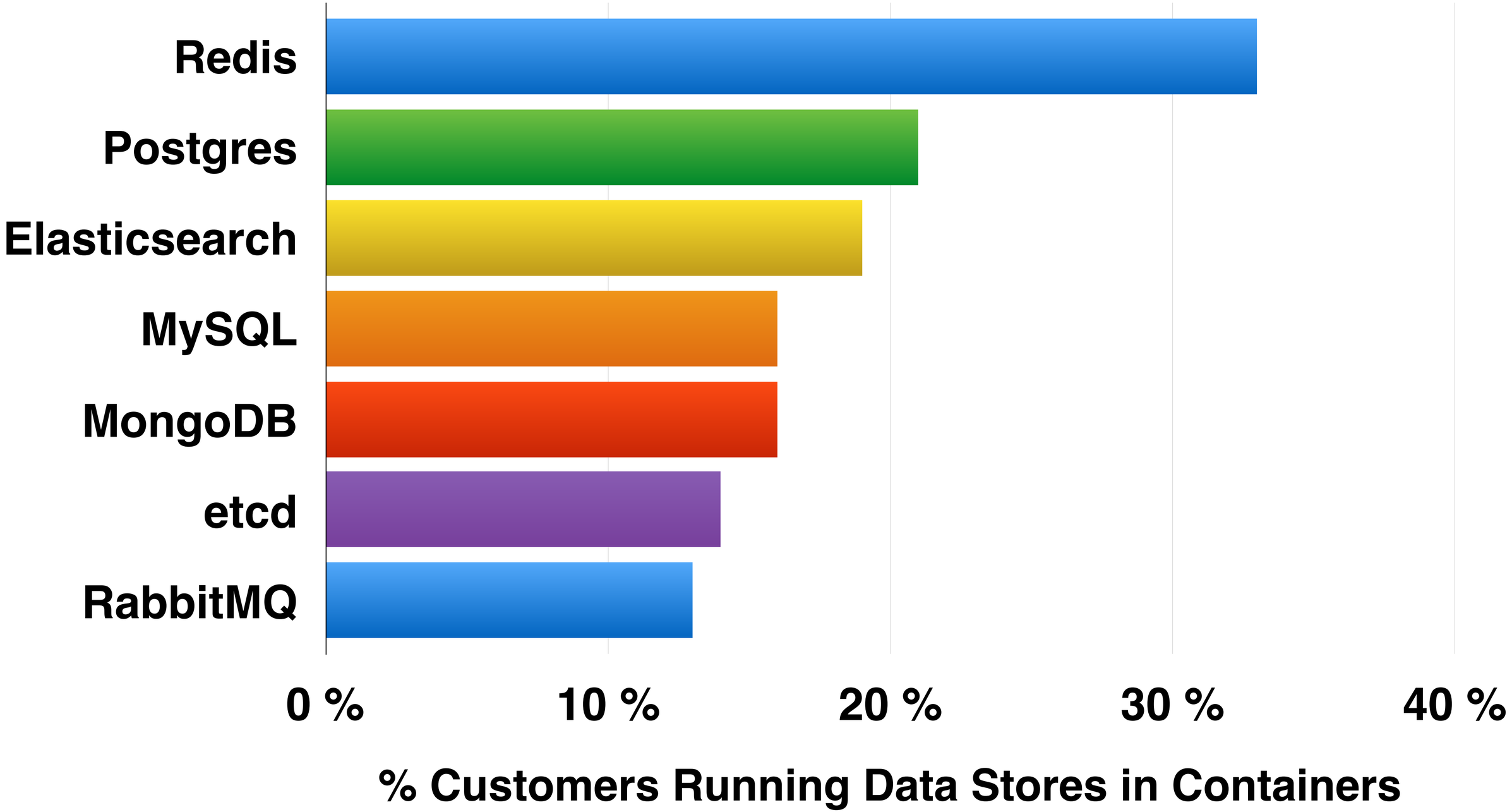
Portion of Hosts Running Docker



Source: Datadog

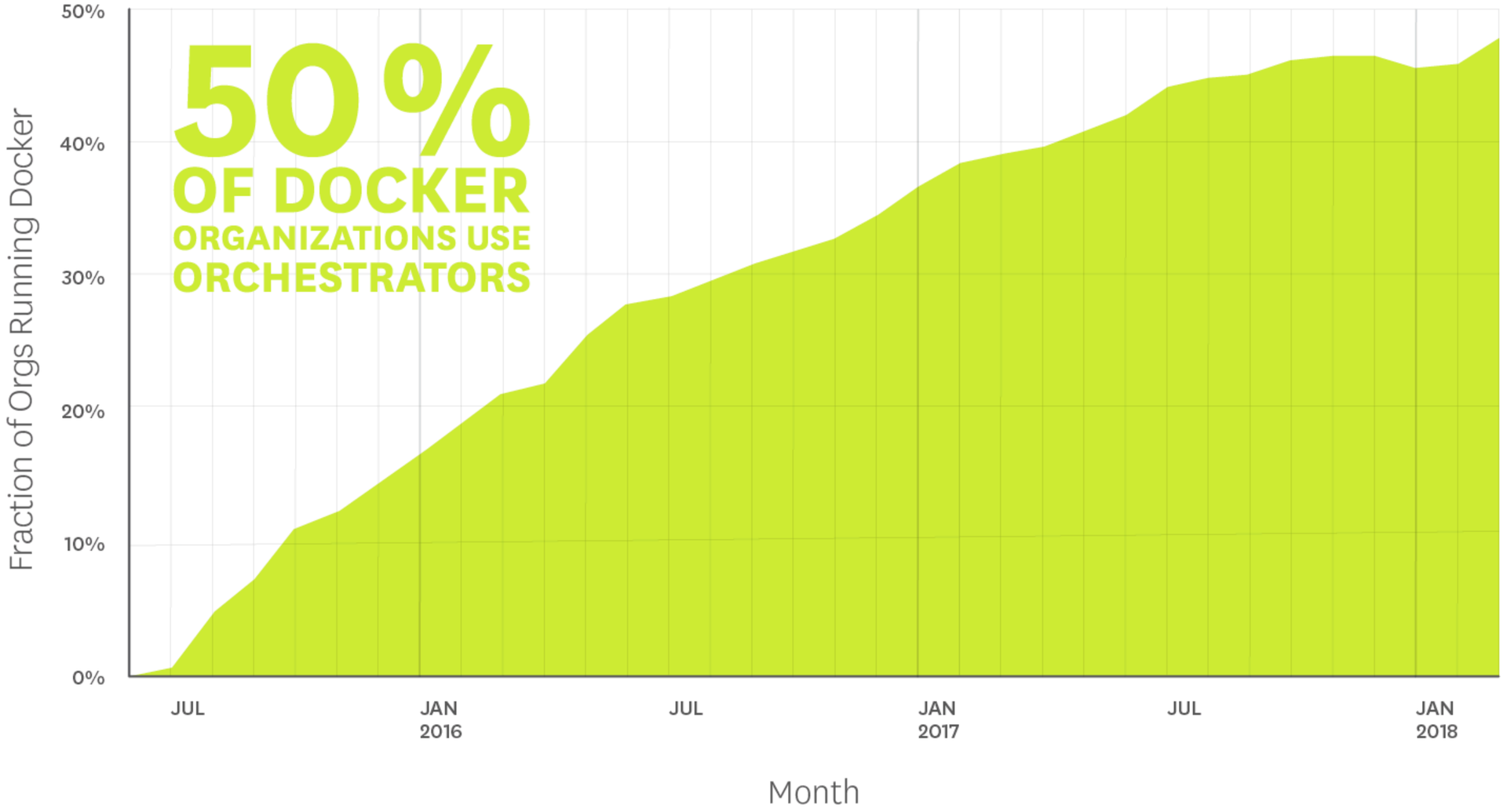
WHAT'S RUNNING?


NGINX
Redis
Postgres
FluentD
Elasticsearch
Mongo
MySQL
etcd
RabbitMQ
HAproxy
0 %
10 %
20 %
30 %
40 %
% Customers Running a Tech in Containers


Redis
Postgres
Elasticsearch
MySQL
MongoDB
etcd
RabbitMQ
0 %
10 %
20 %
30 %
40 %
% Customers Running Data Stores in Containers

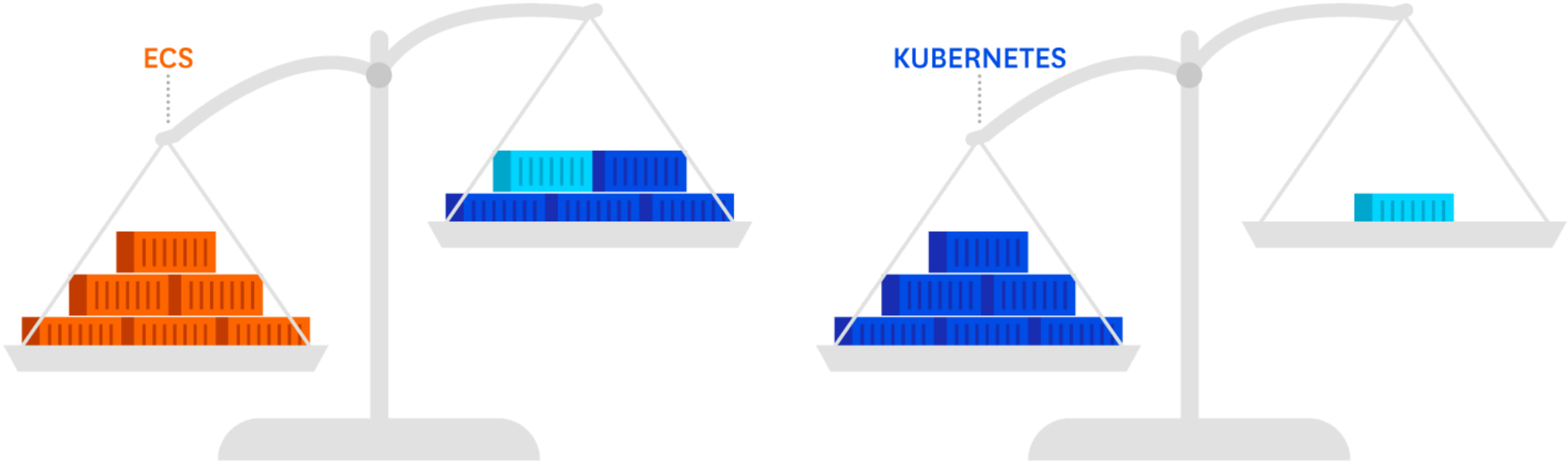
ENTER: ORCHESTRATION

Usage of Orchestrators

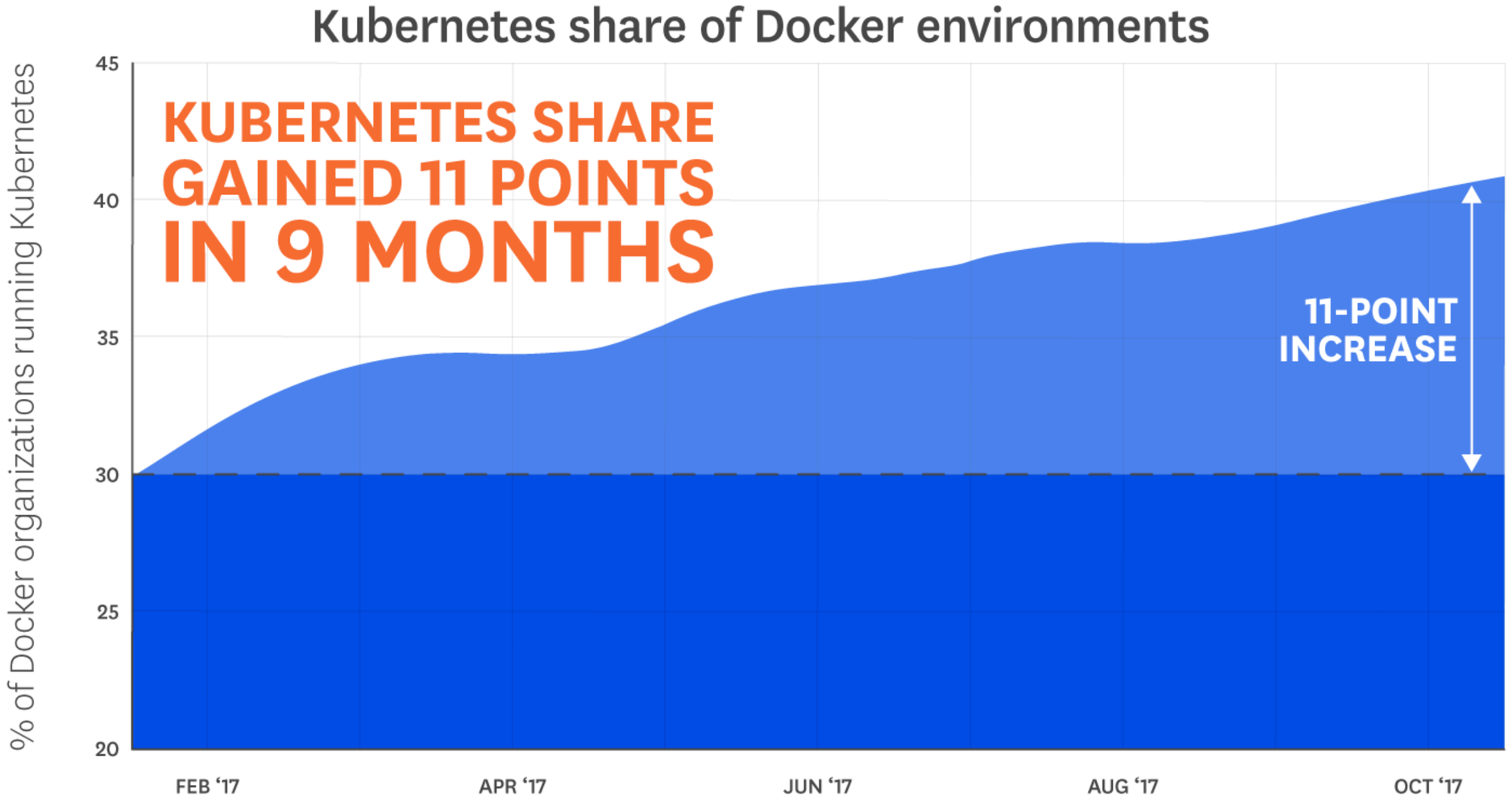


Source: Datadog

ECS PREVAILS IN AWS BUT KUBERNETES LEADS ELSEWHERE

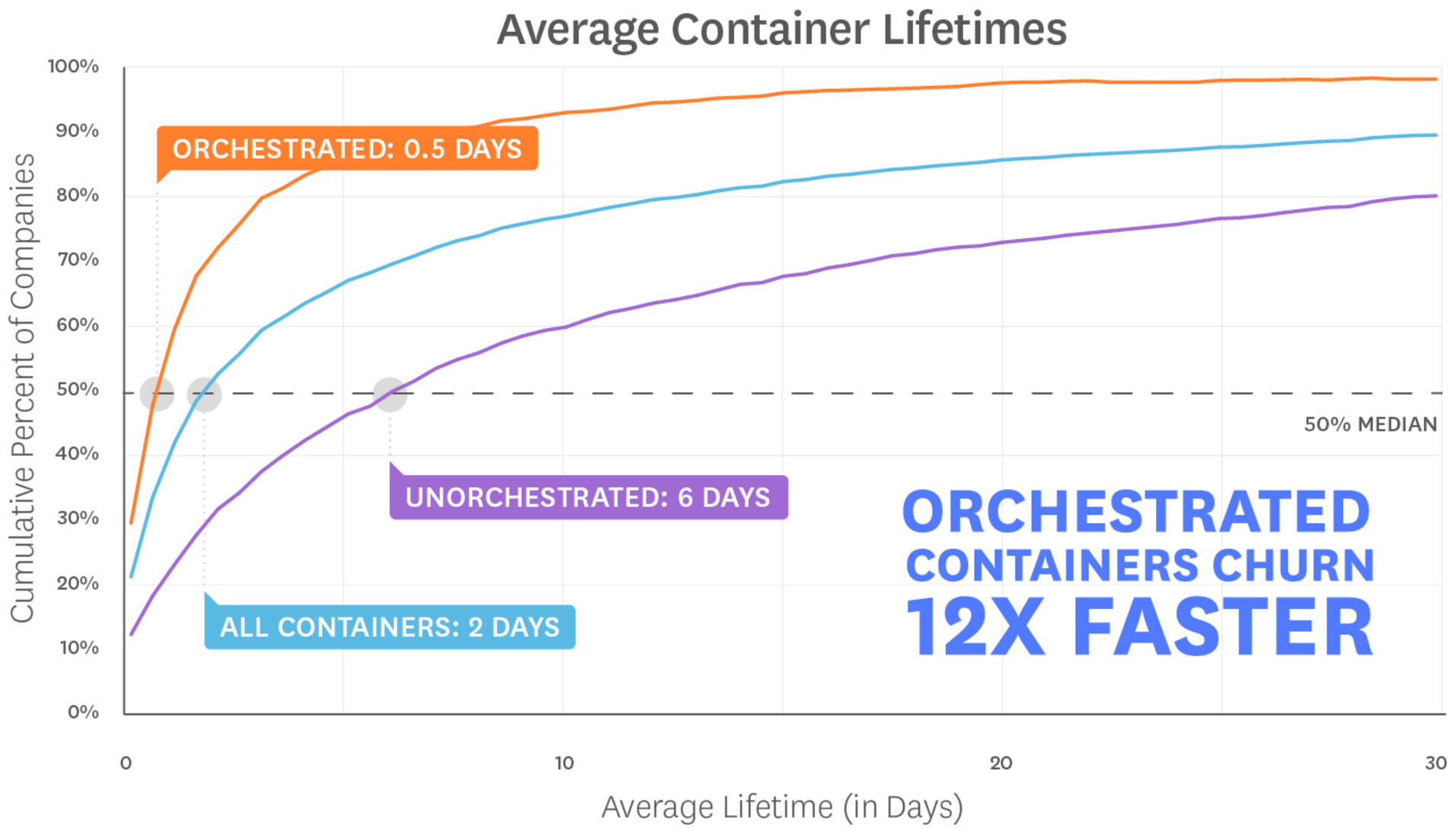


Source: Datadog



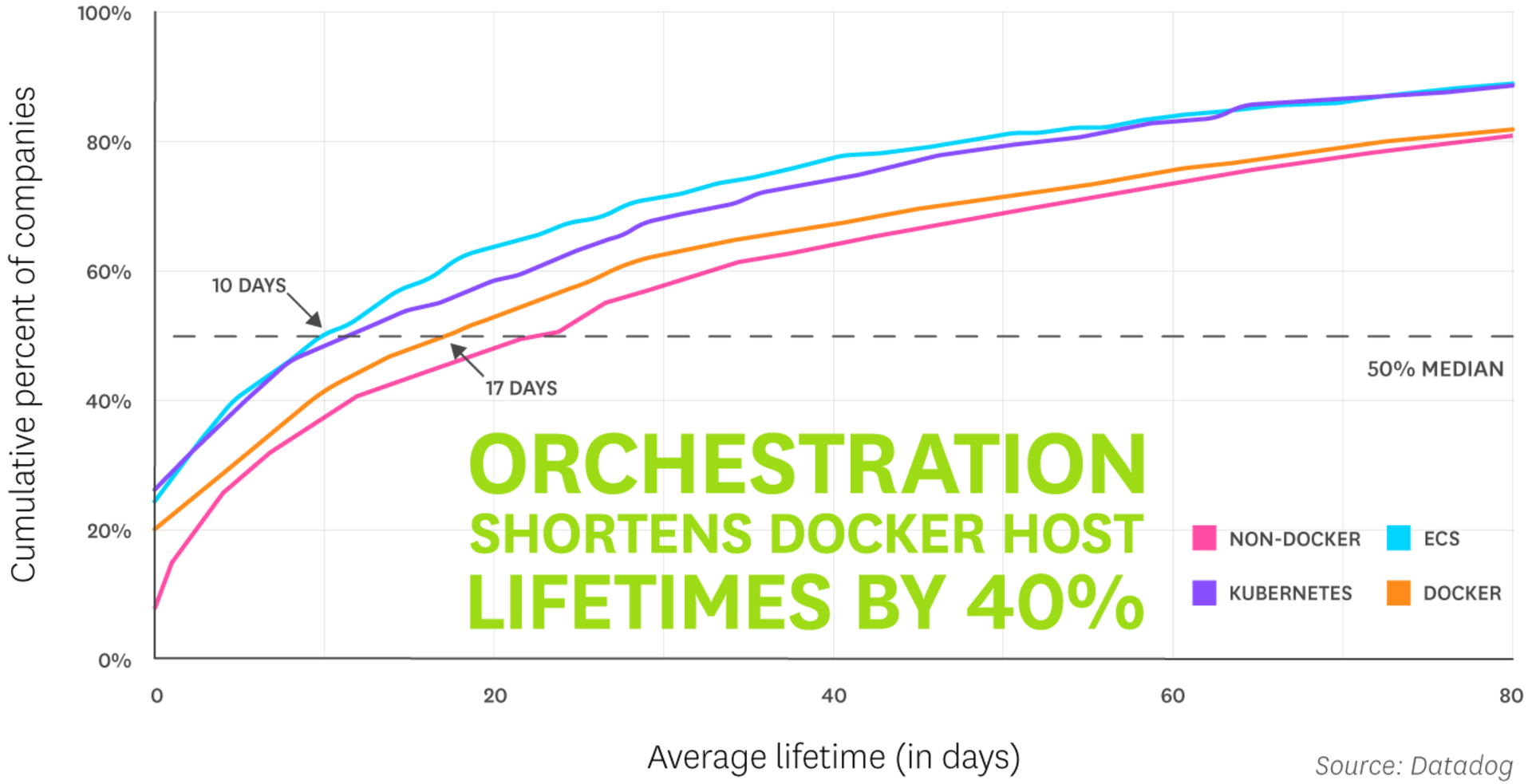
Source: Datadog

NEW CONTAINER: WHO DIS?



Source: Datadog

Average lifetimes of hosts

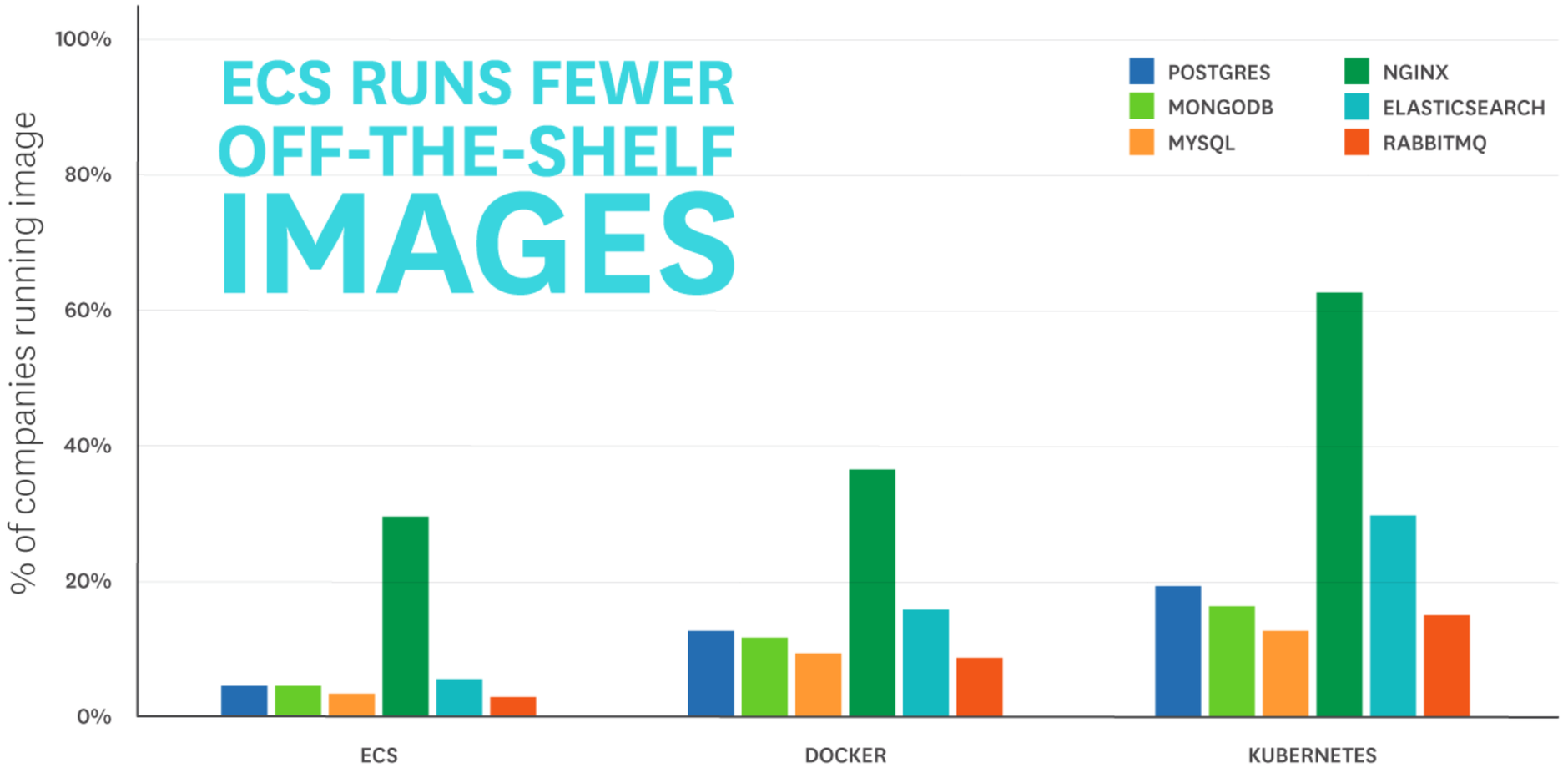


Source: Datadog

WHAT'S RUNNING?

PART 2: THE RUNNINATING

Common container images by orchestrator



Source: Datadog

Companies using tagged image versions

MOST COMPANIES MIX "LATEST" AND VERSIONED IMAGES

74%
Both “latest” and versioned

16%
Only “latest” images

10%
Only versioned images

Source: Datadog

4 QUALITIES OF GOOD METRICS

NOT ALL METRICS ARE EQUAL

1. MUST BE WELL UNDERSTOOD

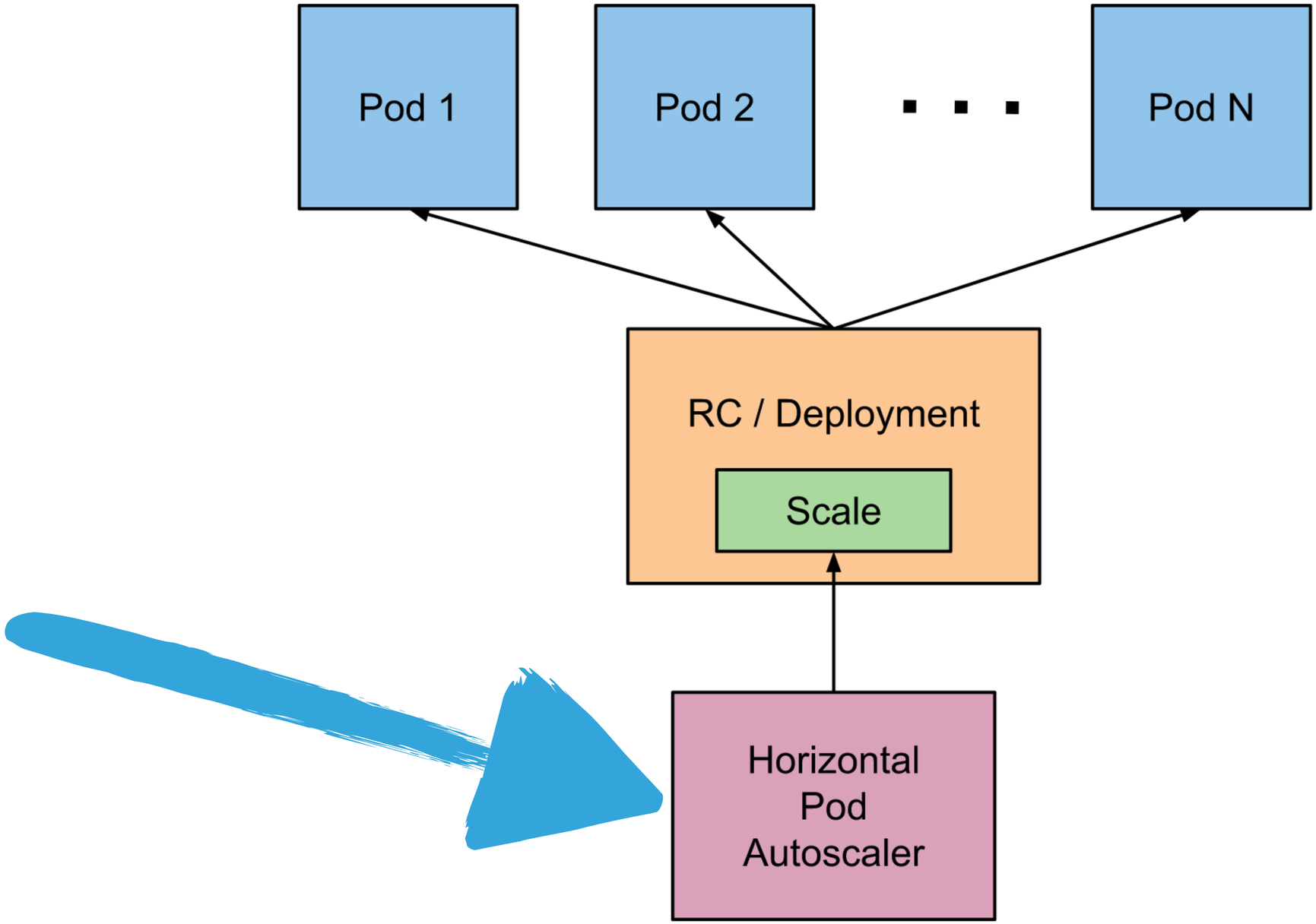
2. SUFFICIENT GRANULARITY

RANK	PARTICIPANT	RESULT
G	Anthony ERVIN USA	21.40
S	Florent MANAUDOU FRA	21.41
B	Nathan ADRIAN USA	21.49
4.	Ben PROUD GBR	21.68
5.	Andrii GOVOROV UKR	21.74
6.	Bruno FRATUS BRA	21.79
6.	Bradley Edward TANDY RSA	21.79
8.	Simonas BILIS LTU	22.08

3. TAGGED & FILTERABLE

4. LONG-LIVED

CUSTOM METRICS & K8S


Pod 1
Pod 2
Pod N
RC / Deployment
Scale
Horizontal
Pod
Autoscaler

1. MUST BE WELL UNDERSTOOD

```
$ kubectl autoscale rs foo --min=5 --max=10 --cpu-percent=80
```

Questions:

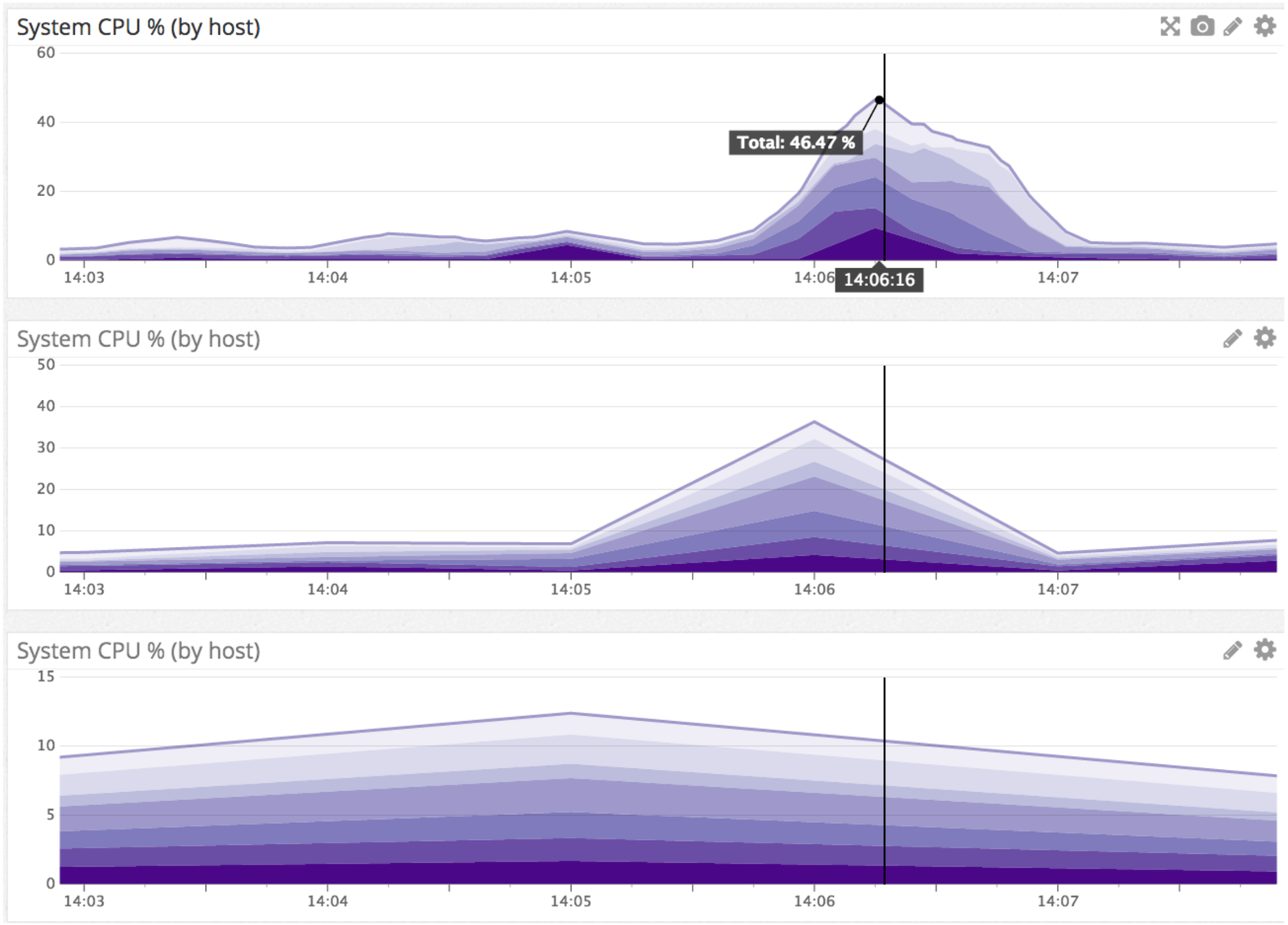
What does cpu percent mean exactly?

Of user-space only? Does that even matter?

Is that on any one instance? Across the pod? Of a single core? Multi-core?

And what about spikes - do those count?

2. SUFFICIENT GRANULARITY



1 second
Peak 46%

1 minute
Peak 36%

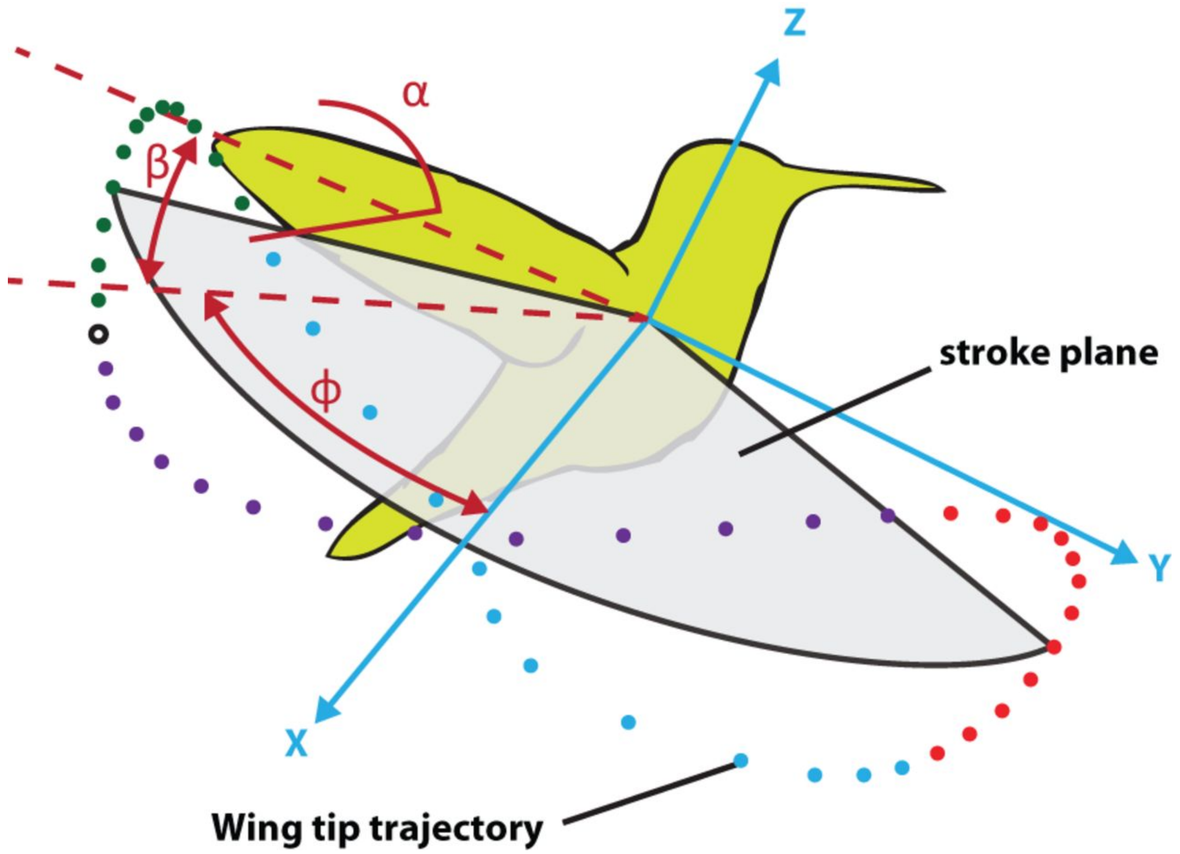
5 minutes
Peak 12%

3. TAGGED & FILTERABLE

“Add more web servers when the frontend is getting busy.”

metric: nginx.net.request_per_s
scope: kube_container_name: frontend

“oh no”



4. LONG-LIVED

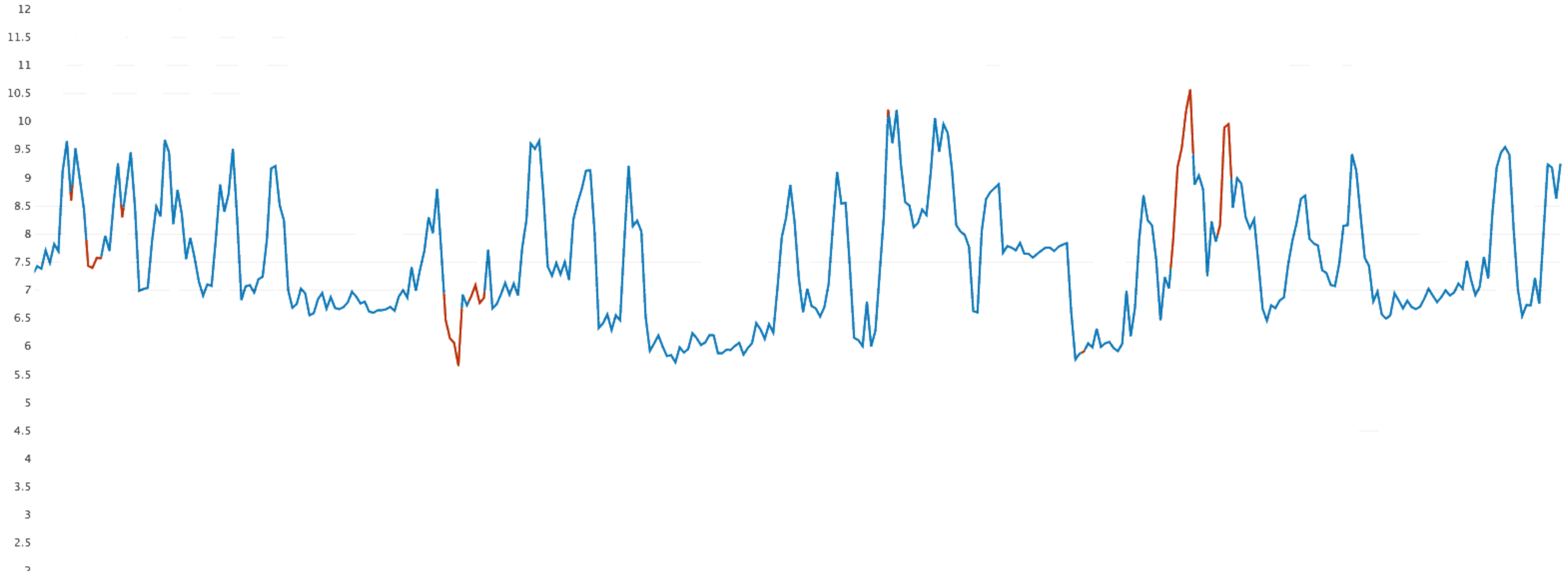
“There always seems to be ten web servers running. Should we increase the maximum number in the set?”

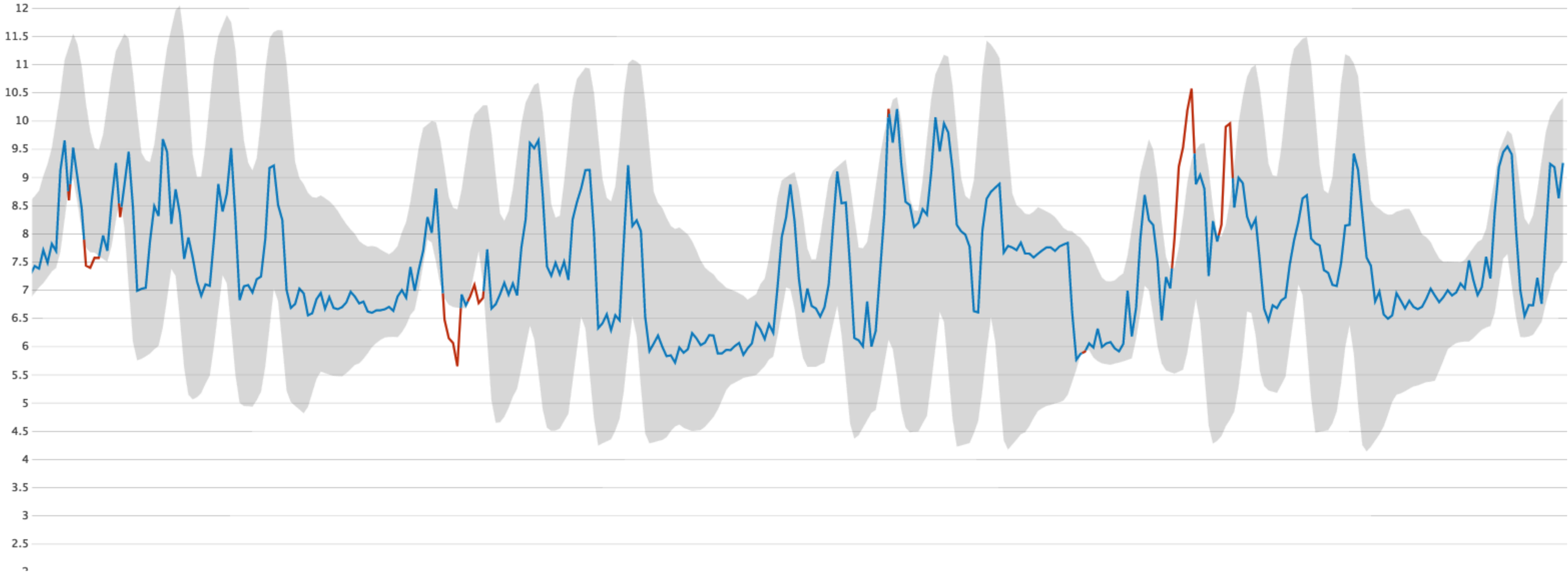
Questions:

What does normal even look like?

Is normal stable over time? What about week-ends?

Did you just notice now? Seriously, how long has it been this way?


12
11.5
11
10.5
10
9.5
9
8.5
8
7.5
7
6.5
6
5.5
5
4.5
4
3.5
3
2.5


12
11.5
11
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10
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8.5
8
7.5
7
6.5
6
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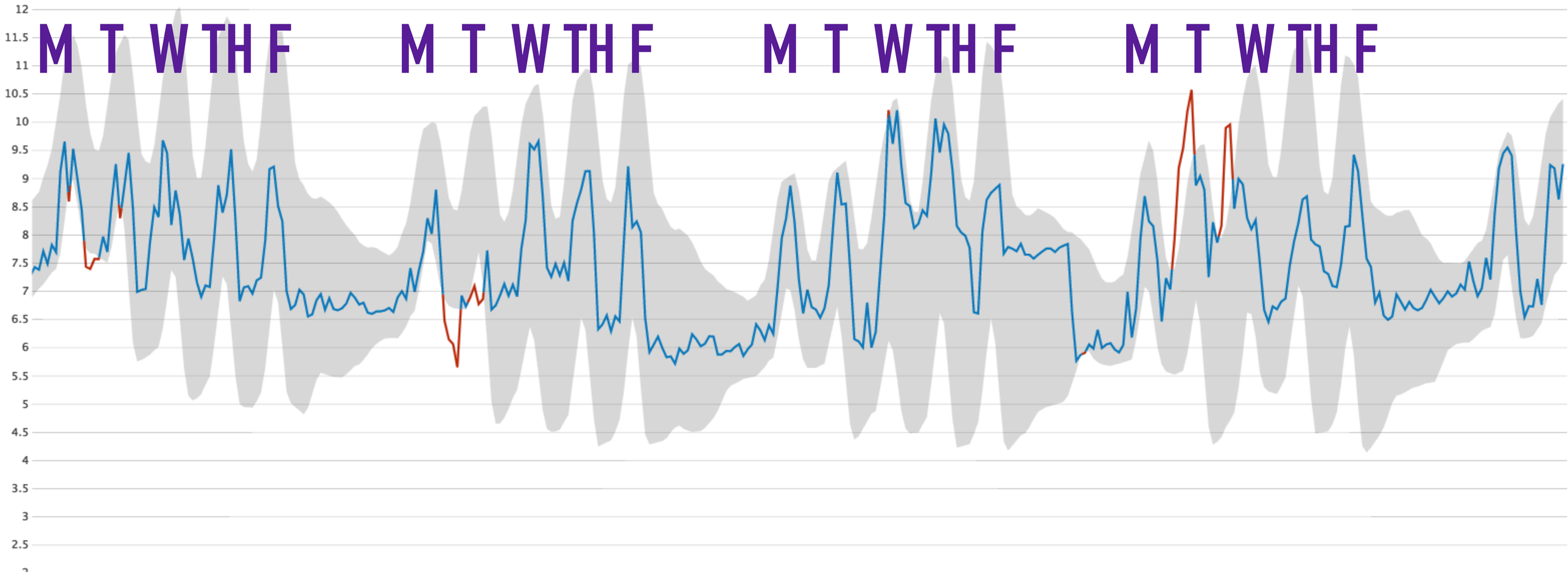
5
4.5
4
3.5
3
2.5


M T W TH F
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M T W TH F
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8.5
8
7.5
7
6.5
6
5.5
5
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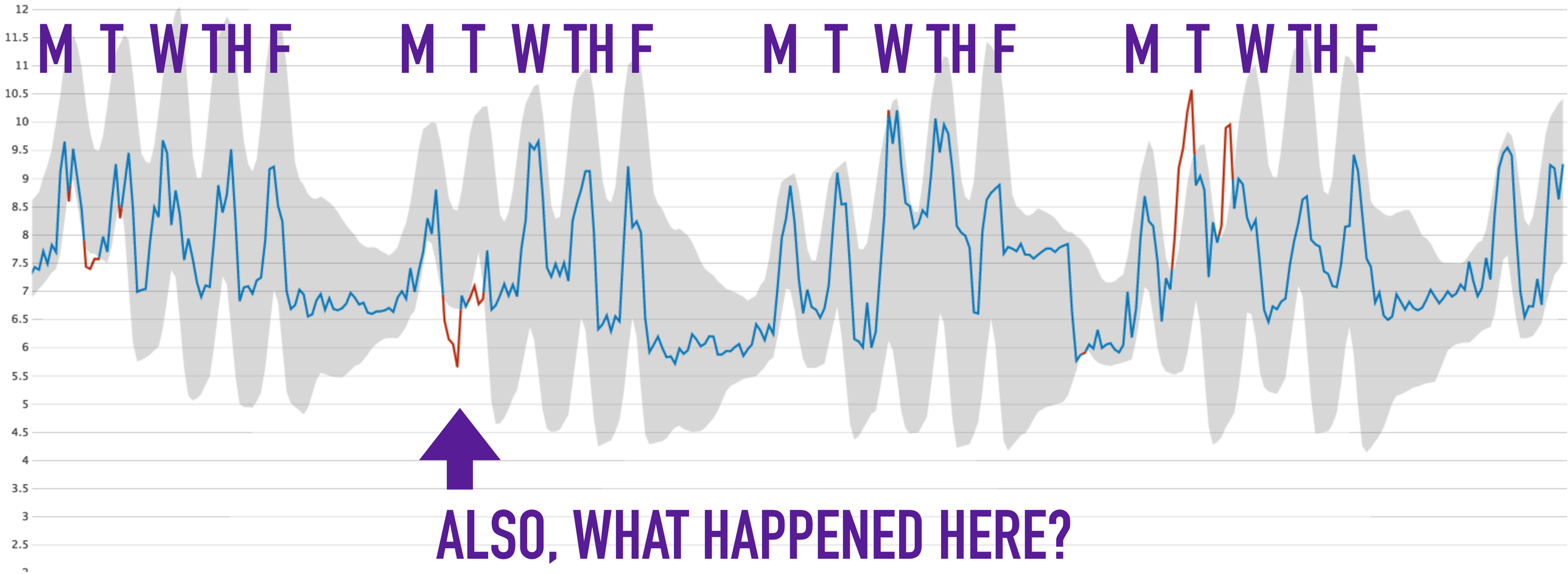
2.5



ALSO, WHAT HAPPENED HERE?

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MERCI !

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